

$$05-) m = 2.500 \text{ kg}$$

$$v_1 = 72 \text{ km/h} \rightarrow v_1 = \frac{72}{3,6} = 20 \text{ m/s}$$

$$v_2 = 90 \text{ km/h} \quad v_2 = \frac{90}{3,6} = 25 \text{ m/s}$$

$$E_{c0} = \frac{mv_1^2}{2} = \frac{1500 \cdot 20^2}{2} = \frac{1500 \cdot 20^2}{2} = 300.000 \text{ J}$$

$$E_{cf} = \frac{mv_2^2}{2} = \frac{1500 \cdot 25^2}{2} = \frac{1500 \cdot 25 \cdot 25}{2} = 468.750 \text{ J}$$

$$E_{cf} - E_{c0} = \Delta E \rightarrow \text{diferença de energia cinética}$$

$$\frac{1500 \cdot 25^2}{2} - \frac{1500 \cdot (20^2)}{2} = \Delta E$$

$$\Delta E = \frac{1500 (25^2 - 20^2)}{2} = \frac{1500 (25 - 20)(25 + 20)}{2}$$

$$\Delta E = 168.750 \text{ J} \sim 1,7 \cdot 10^5 \text{ J}$$

alternativa "d"

06-) $m = 50 \text{ kg}$; $d = 30 \text{ cm} \rightarrow d = 0,3 \text{ m}$

$t_1 \rightarrow W = m \cdot g \cdot h = 50 \cdot 10 \cdot 0,3 = 150 \text{ J}$

em t_2 foram gastos 450 J

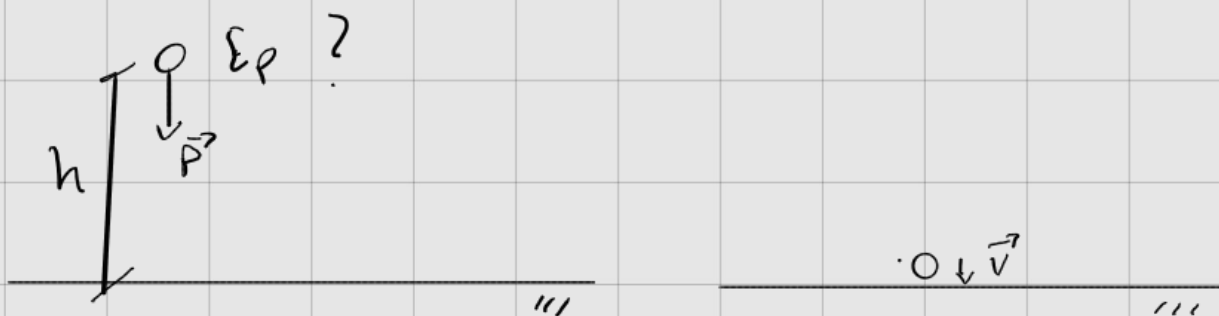
$\hookrightarrow 450 - 150 = 300 \text{ J} \rightarrow \text{tem } 300 \text{ J "sobrando"}$

$W_3 = m \cdot g \cdot h$

~~300~~ $= 50 \cdot 10 \cdot h$

$h = \frac{3}{5} = 60 \text{ cm}$ alternativa "b"

09-)



Se dobrar $v \rightarrow E_c$ é multiplicada por 4

por conservação: E_p tem que ser $\times 4$ também

$E_p = m \cdot g \cdot h \rightarrow E_p = m \cdot g \cdot 4h$

alternativa "c"

$$10-) \quad \bar{E}_{m_0} = E_m$$

$$\cancel{m \cdot g \cdot h} + \frac{m v_0^2}{2} = m \cdot g \cdot h + E_c$$

$$\frac{m v_0^2}{2} - m \cdot g \cdot h = E_c$$

$$E_c = m \left(\frac{v_0^2}{2} - g \cdot h \right)$$

$$E_c = \frac{1}{2} \left(\frac{10^2}{2} - 10 \cdot 2 \right) = 15 \text{ J}$$

"alternativa c"

13-)

○ ↓ $\vec{v}$

_____ "

$$\bar{E}_p = m \cdot g \cdot h = 5 \cdot 10 \cdot 4 = 200 \text{ J}$$

$$\bar{E}_c = \frac{1}{2} m \cdot v^2 = \frac{5 \cdot 6^2}{2} = 90 \text{ J}$$

"alternativa a"

$$\Delta E_m = 110 \text{ J}$$

} 90 J "perdido"

caso contrário a

velocidade seria maior

14-) $m = 200 \text{ g} \rightarrow 0,2 \text{ Kg}$; $K = 2000 \frac{\text{N}}{\text{m}}$



$x = 4 \text{ cm} = 0,04 \text{ m}$

$$E_{pe} = \frac{K \cdot x^2}{2} = \frac{2000 \cdot (0,04)^2}{2} = \frac{2000(4 \cdot 10^{-2})^2}{2}$$

$$E_{pe} = \frac{2 \cdot 10^3 \cdot 16 \cdot 10^{-4}}{2} = 1,6 \text{ J}$$

0,6 J e' dissipado

$\hookrightarrow E_{p_{\text{grav}}} = 1,6 - 0,6 = 1 \text{ J}$

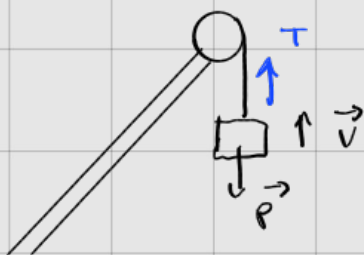
$$E_{p_{\text{grav}}} = m \cdot g \cdot h$$

$$1 = 0,2 \cdot 10 \cdot h$$

$$h = \frac{1}{2} = 0,5 \text{ m}$$

"alternativa b"

15-)



$$m = 10 \text{ kg}$$

$$v = 5 \text{ m/s}$$

$T = m \cdot g$ força MINIMA (peso) para erguer

$$T = 10 \cdot 10 = 100 \text{ N}$$

$$P = \frac{W}{\Delta t} = \frac{F \cdot d}{\Delta t} \quad \text{velocidade média}$$

como v é constante $v_{\text{média}} = v$

$$P = F \cdot v = 100 \cdot 5 = 500 \text{ W}$$

"alternativa b"

$$16-) \quad m = \rho Q = \frac{\text{massa}}{s} \quad \text{densidade} \cdot \text{vácuo}$$

↳

$$P_p = \rho Q g h$$

$$P_p = \rho \cdot Q \cdot g \cdot h = 1000 \cdot 690 \cdot 10 \cdot 118,4 = 816,96 \text{ MW}$$

$$\Delta P = (816,96 - 14 \cdot 20) \cdot 10^6$$

$$\Delta P = 116,96 \cdot 10^6 \text{ W}$$

- "alternativa c"

17) 18% de rendimento

$$3 \times 6 \text{ m} \rightarrow A = 18 \text{ m}^2$$

$$I = 1000 \frac{\text{W}}{\text{m}^2} \rightarrow \text{para } 18 \text{ m}^2 \rightarrow$$

$$P = 18000 \text{ W} = 18000 \frac{\text{J}}{\text{s}}$$

$$P_{\text{útil}} = 0,18 \cdot 18000 = 3240 \text{ W} = 3,24 \text{ kW}$$

↳ 18%

↳ obs: isso é em 1h

$$\hookrightarrow 3,24 \cdot 6 \rightarrow 19,44 \text{ kW} \cdot \text{h} / \text{dia}$$

$$\text{em 30 dias} \rightarrow 19,44 \cdot 30 = 583,2 \text{ kWh}$$

"alternativa c"

$$18-) P = 3000 \text{ W}$$

durante 8h / dia

$$\hookrightarrow 3000 \cdot 8 = 24.000 \text{ Wh}$$

durante 30 dias

$$\hookrightarrow 24000 \cdot 30 = 720.000 \text{ Wh} = 720 \text{ kWh}$$

"alternativa a"

$$19-) \quad m = 200 \text{ kg}$$

$$A = 9 \text{ m}^2$$

$$I = 1000 \frac{\text{W}}{\text{m}^2}$$

$$\eta = 30\%$$

$$v = 108 \text{ km/h} = 30 \frac{\text{m}}{\text{s}}$$

$$\begin{array}{rcl} 1 \text{ m}^2 & \text{---} & 1000 \text{ W} \\ 9 \text{ m}^2 & \text{---} & x \end{array} \quad \left. \vphantom{\begin{array}{rcl} 1 \text{ m}^2 & \text{---} & 1000 \text{ W} \\ 9 \text{ m}^2 & \text{---} & x \end{array}} \right\}$$

$$x = 9 \cdot 1000$$

$$x = 9000 \text{ W}$$

$$\begin{array}{rcl} 100\% & \text{---} & 9000 \\ 30\% & \text{---} & x \end{array} \quad \left. \vphantom{\begin{array}{rcl} 100\% & \text{---} & 9000 \\ 30\% & \text{---} & x \end{array}} \right\}$$

$$x = 30\% \cdot 9000$$

$$x = 2700 \text{ W}$$

$$E_0 = \frac{m \cdot v^2}{2} = \frac{200 \cdot 30^2}{2} = 900 \cdot 100 = 90000 \text{ J}$$

$$E = P \cdot \Delta t$$

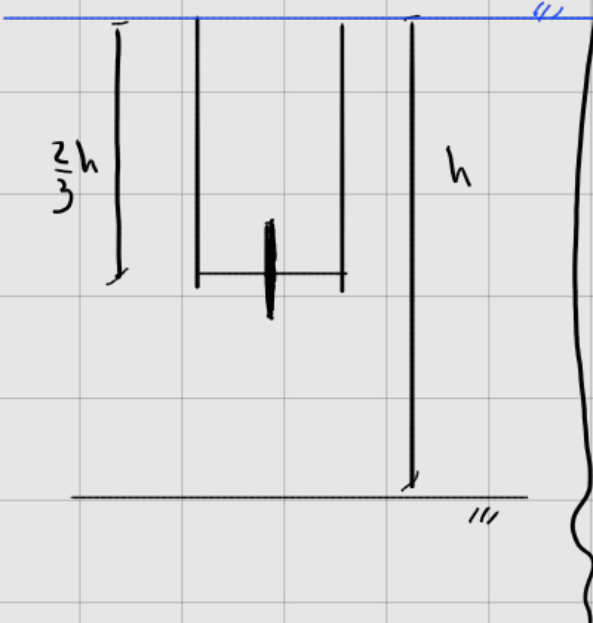
$$90000 = 2700 \cdot \Delta t$$

$$\Delta t = \frac{90000}{2700} = 33,3 \text{ s}$$

20.) $m = 30 \text{ g} = 30 \cdot 10^{-3} \text{ kg}$

$h = 410 \text{ mm}$

referência



$$E_{p_{\max}} = m \cdot g \cdot H = m \cdot g \cdot h$$

$$E_{p_{\max}} = 30 \cdot 10^{-3} \cdot 10 \cdot \frac{2}{3} \cdot 410 \cdot 10^{-3}$$

$$E_{p_{\max}} = 10^{-3} \cdot 820 \cdot 10^{-3}$$

$$E_{p_{\max}} = 10^{-3} \cdot 820 \cdot 10^{-3} = 820 \cdot 10^{-6} = 8,2 \cdot 10^{-2} \text{ J}$$

"alternativa b"